The Fairness-Performance Pareto Frontier: How to Design Less Discriminatory Algorithms?

MIEKE WILMS, Zurich University of Applied Sciences / University of Zurich, Switzerland
CHRISTOPH HEITZ, Zurich University of Applied Sciences, Switzerland

In our recent publication we derive an optimality theorem for binary data-based decision systems which characterizes the Pareto-optimal systems in the performance-fairness space, for a wide range of group fairness scores and arbitrary performance utility functions [11]. The theorem gives a theoretical boundary for the Pareto frontier that cannot be exceeded by any algorithm, and proves that these solutions are deterministic group-specific threshold rules.

We recapitulate the most important findings of this paper under the perspective of their *legal* implications. A decision system to be qualified as legally non-discriminatory needs to pass the so-called *proportionality test* which (among other questions) requires a proof that the goal of the decision maker could not have been achieved with less discriminatory means. Technically, this implies that the decision system is Pareto-optimal in the performance-fairness space, evaluated with a suitable fairness metric. We show that the optimality theorem of [11] helps to deal with the legal question of the proportionality test, but at the same time leads to puzzling questions with respect to the legal concepts of direct and indirect discrimination.

Keywords: decision systems, group fairness, Pareto frontier, multi-objective optimization, fairness-utility tradeoff

Reference Format:
Mieke Wilms and Christoph Heitz. 2026. The Fairness-Performance Pareto Frontier: How to Design Less Discriminatory Algorithms?. In *Proceedings of Fifth European Conference on Algorithmic Fairness (ECAF'26).* Proceedings of Machine Learning Research, 5 pages.

1 Introduction

Algorithmic decision systems are widely used nowadays. They are designed to achieve the goals of a decision maker (DM), but also affect the lives of the decision subjects (DS), raising concerns about discrimination, unfairness and social injustice with respect to social salient groups. [11] addresses the tension between performance (DM perspective) and fairness/non-discrimination (societal perspective) by providing a theorem for Pareto-optimality.

Finding Pareto-optimal decision systems in the two-dimensional performance-fairness space is a relatively new research area [7, 8, 10, 12]. While posing a challenging ML problem by itself, the Pareto front is also extremely important from a legal perspective. According to most discrimination laws, disparate treatment is allowed if justified by a legitimate aim (e.g. maximizing profit) and if additional requirements are met (collectively called the *proportionality test* in European law [9]). One of the requirements is that it must be impossible to achieve the same performance with *less* disparity, which technically means that the system is Pareto-optimal.

Authors' Contact Information: Mieke Wilms, Zurich University of Applied Sciences / University of Zurich, Zurich, Switzerland, mieke.wilms@zhaw.ch; Christoph Heitz, Zurich University of Applied Sciences, Zurich, Switzerland, christoph.heitz@zhaw.ch.

ECAF'26, September 02–September 04, 2026, Ghent, BE

We show in [11] that, for binary classifiers $\vec{x} \mapsto D \in \{0,1\}$, with $\vec{x}$ a feature vector, Pareto optimal solutions are characterized by *group-specific threshold rules* applied to the Bayes-optimal probability $p(\vec{x}) = P[Y=1|\vec{x}]$, where $Y = \{0,1\}$ is a decision-critical variable that is unknown at the time of decision making, and perfect decisions are characterized by $D = Y$. This result is valid for the whole Pareto front and holds for a wide range of fairness metrics and for arbitrary utility functions of DM. Most notably, the optimal decision rule may be an *upper-bound* threshold rule, i.e. $D = 1$ if $p < t$ (with threshold $t \in [0,1]$). This is opposed to classical optimality theorems [4, 5] which lead to lower-bound decision rules ($D = 1$ if $p > t$). We show that upper-bound threshold rules may occur if the DS utility function, used for modeling benefit or harm of decisions, meets certain conditions.

The optimality theorem characterizes the Pareto-optimal decision systems in the space of *all possible* binary decision systems based on $\vec{x}$, which allow implementation of group-specific thresholds. Hence, it is independent of how these systems are technically implemented and therefore includes all pre-processing, in-processing and post-processing approaches. We thus conclude that any decision system that is designed to optimize both performance and fairness will tend to replicate these optimal group-dependent threshold rules, as far as its technical design allows. As evidence, we show that a well-known in-processing method for learning Pareto-optimal solutions actually does implement group-specific thresholds, even without using the sensitive attribute a.

Formal description

We model a binary decision system as mapping $f : \vec{x} \mapsto D$ from observable feature vector $\vec{x}$ to decision $D \in \{0,1\}$, where f is applied repeatedly to many individuals of a population. A decision critical binary variable Y is not known at the time of decision making. The DM is interested in maximizing their expected utility $E(U)$ across all individuals, where U is a function of both D and Y, and $E(U)$ is maximum for $D = Y$. For instance, a bank (DM) decides whether a loan application (DS) should be accepted (D=1) or declined (D=0). Ideally, the application is accepted when the applicant does not default (Y=1) and declined when the applicant defaults (Y=0). However, since Y is unknown, the bank implements a decision rule $f : \vec{x} \mapsto D$ optimized for $E(U)$.

For modeling the impact of the decisions on the DS, we use the utility-framework of [3]. The average benefit of a decision for a member of group a is given by the expectation value $E[V|a]$ of a DS utility V, also depending on both D and Y. A fairness score FS is a function of the different $E[V|a]$ for the different groups a. We include expected values conditioned on Y ($E[V|Y,a]$) or on D ($E[V|D,a]$). This framework includes, as special cases, all classical confusion-based fairness metrics[1]. Furthermore, it allows implementation of different principles of justice [6].

2 Characterizing the Pareto frontier of algorithmic decision systems

Pareto-optimal decision rules

In [11], we provide a step-wise approach to characterize decision rules that are Pareto-optimal with respect to fairness (FS) and performance ($E[U]$). This results in the following optimality theorem.

THEOREM 2.1. *Given a DM utility function, a (possibly group-specific) DS utility function, and a fairness score FS which is a function of the (possibly conditioned) expected DS utilities. We consider the set of all possible decision rules $\mathscr{D} : \vec{x} \mapsto D$ including those that allow independent decision rules for different groups $a \in A$.*

[1]Classical confusion-based fairness metrics as *false positive/negative rate* or *positive/negative predictive value* form the basis of fairness criteria such as independence, separation and sufficiency [1].

Then it holds that any decision rule $d \in \mathscr{D}$ which is Pareto-optimal with respect to $(E[U], FS)$ implements group-specific deterministic threshold rules applied to $p = P[Y = 1|\vec{x}]$ for all groups $a \in A$. For each group, this may be either a lower-bound (lb) or an upper-bound (ub) threshold rule with threshold $t_a \in [0,1]$:

$$d_{lb} = \begin{cases} 1 & \text{for } p \geq t_a \\ 0 & \text{otherwise} \end{cases} \qquad \text{or} \qquad d_{ub} = \begin{cases} 1 & \text{for } p < t_a \\ 0 & \text{otherwise} \end{cases} .$$

In the case of two groups ($a \in \{0,1\}$), the following four combinations are candidates for Pareto-optimal decision rules: *lb-lb* (lb thresholds for both groups), *lb-ub* (lb threshold for group 0, and ub for group 1), *ub-lb* (vice versa) and *ub-ub* (ub thresholds for both groups).

Theoretical example: Pareto-optimal decision rules may be upper-bound threshold rules

We consider a population that consists of two groups, labeled 0 and 1 (refer to [11], Section 4, for details). As fairness score, we chose $FS = |E[V|a = 0] - E[V|a = 1]|$. Figures 1 a.-c. illustrate the candidate decision rules (grey points). The lower-right outer bound forms the Pareto frontier as the objective is to maximize performance ($E[U]$) and minimize FS. We indicate the Pareto frontier for the four combinations (*lb-lb* / *lb-ub* / *ub-lb* / *ub-ub*) separately to visualize which types of decision rules constitute the outer bound.

In Figure 1a., we use $E[V|a] = P[D = 1|a]$ (*selection rate*), which, for $FS = 0$, corresponds to *demographic parity* (point B). Fig. 1a. shows that, for this fairness metric, all Pareto-optimal decision rules are lb-lb threshold rules. This was already proven by Hardt et al. [5] and Corbett-Davies et al. [4] for point B, and we show that it actually holds for the entire Pareto frontier.

Figures 1b. and c. are derived analogously, however with a slightly different DS utility function, resulting in different values $E[V|a]$. Here, the Pareto-frontier includes lb-ub threshold rules. Specifically, for group 1, individuals with low probabilities p are preferred over those with high probabilities. This outcome may appear counterintuitive, but it is consistent with previous findings of Baumann et al. [2], showing similar results when enforcing sufficiency constraints.

Empirical example: Black-box algorithms may produce group-dependent threshold rules

Our theoretical results suggest that any decision system designed for optimizing both fairness and performance will replicate group-dependent threshold rules, as far as its technical design allows. As validation, we evaluate the in-processing algorithm PF-SMG (*Pareto front stochastic multi-gradient* [7]) which determines Pareto-optimal classifiers, without using the sensitive attribute a. Following our approach, we use Logistic Regression estimates $\hat{p}_{LR}$ as approximations of $p = P[Y = 1|\vec{x}]$ and derive a second Pareto frontier by evaluating all combinations of group-dependent threshold rules applied to $\hat{p}_{LR}$ (refer to [11], Section 5, for details on experimental set-up).

Figures 1 d.-f. display the results of applying both methodologies on the Adult Income dataset. Figure 1 d. shows that our Pareto frontier clearly dominates that of [7][2]. More important, however, is the analysis of the proposed decision rules, by plotting the fraction of individuals with $D = 1$ against $\hat{p}_{LR}$. In Figures 1 e. and f., the decision rules for two different levels of FS are tested. As predicted by our theoretical analysis, the decision rules of PF-SMG approximate the group-specific threshold rules suggested by our approach.

[2]This might not seem surprising as our approach uses the sensitive attribute a during decision making, by implementing group-dependent decision rules (though not for estimating $\hat{p}_{LR}$), while the decision rules of [7] do not have access to a.

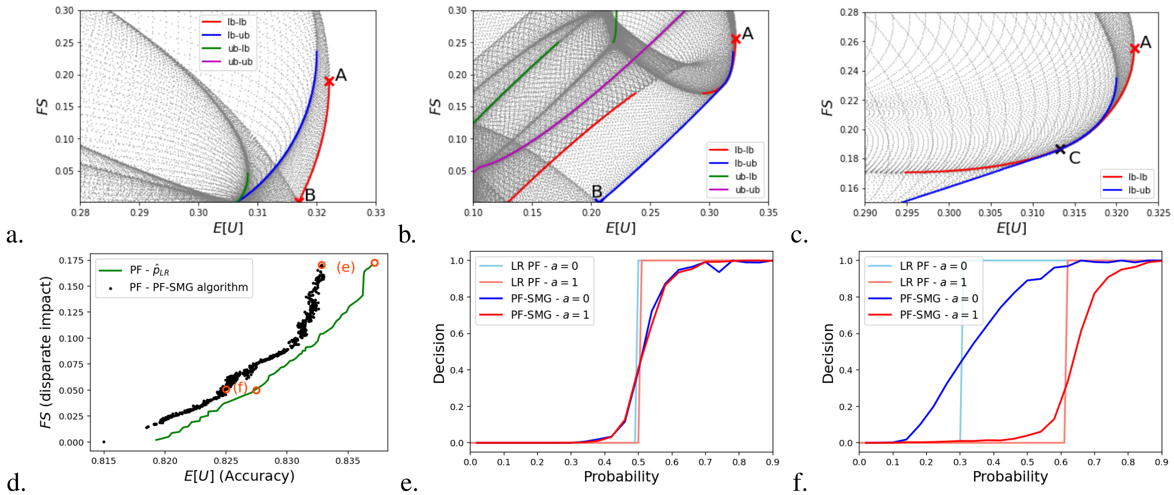

Fig. 1. **(a.-c.)** The Pareto frontier for different DS utility functions. For each combination (*lb-lb* / *lb-ub* / *ub-lb* / *ub-ub*), candidate decision rules (grey points) and their associated Pareto frontiers (colored lines) are plotted. The overall Pareto frontier is defined as the set of all candidate rules. Fig. c. is a section of Fig. b.
(d.-f.) Comparison of the Pareto frontier (PF) obtained by the PF-SMG algorithm and our approach (d.) and the implemented decision rules with respect to Logistic Regression estimates $\hat{p}_{LR}$ of $P[Y=1|\vec{x}]$ (e. and f.).

3 Conclusion

We derived a theorem on Pareto-optimality of binary decision systems with respect to $(E(U), FS)$, valid for a wide range of fairness scores FS. This theorem includes classical optimality results for enforcing fairness constraints [2, 4, 5], but extends them to the complete Pareto front, from fully ignoring fairness to fully establishing fairness. We find that Pareto-optimal solutions are characterized by deterministic group-specific threshold rules. However, depending on the DS utility function, this may include upper-bound threshold rules. The optimality theorem holds for the set of *all possible* decision systems based on a given feature vector $\vec{x}$ including those that allow group-dependent decisions. Thus, any technical approach for realizing a Pareto optimal decision system will tend to implement such deterministic group-specific threshold rules, as far as its technical design allows. Section 2 shows a concrete example of an in-processing method and proves that decision rules $\vec{x} \mapsto D(\vec{x})$ may produce group-specific thresholds even if the sensitive attribute a is not used at all.

Our results not only lead to a simple way of finding the location of the Pareto frontier by evaluating the set of possible candidates for Pareto-optimal solutions. They also help in dealing with an important legal question arising in discrimination law (proportionality test). However, at the same time new legal questions arise: If a legally non-discriminating algorithm requires a Pareto-optimal decision rule, it must necessarily implement a group-specific decision rule, which would be deemed direct discrimination. In other words: Avoiding indirect discrimination leads to direct discrimination. Further interdisciplinary research seems needed to solve this apparent legal contradiction.

Acknowledgments

This work is supported by a grant of the *Digitalisierungsinitiative der Zürcher Hochschulen*.

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